

Richard Harris

Senior Database Engineer | Database Architect

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Profile

Senior Database Engineer with **12+** years designing and optimizing secure, high-performance data platforms for **healthcare** and **finance**. Expert in **Oracle, SQL Server, PostgreSQL, AWS, Azure, Snowflake**, and data engineering pipelines (**Airflow, ADF, Glue**). Proven success in **cloud** migration, **automation**, and **performance** tuning to drive operational efficiency, business resilience, and regulatory compliance (**HIPAA, PCI-DSS, SOC 2**).

Education

Master of Science in Computer Science,
University of Texas at Dallas

2010 – 2012 | Dallas, TX

Skills

Scripting & Languages

SQL · PL/SQL · T-SQL · Python · PowerShell · Shell Scripting

Cloud Platforms

AWS (RDS, Aurora, Redshift, Glue, S3, IAM, KMS, DynamoDB, Secrets Manager)

Azure (SQL, Managed Instance, Data Lake, Data Lake Gen2, Synapse, Data Factory, Monitor, Key Vault, AD, Cosmos DB)

Data Governance & Cataloging Tools

Microsoft Purview · Azure Data Catalog

Automation & Infrastructure as Code (IaC)

Terraform · Liquibase · GitHub Actions · Azure DevOps · ARM Templates · Bicep

Monitoring & Observability

Prometheus · Grafana · OpenTelemetry · SolarWinds DPA · Azure Monitor · Query Store · Redgate · OEM (Oracle Enterprise Manager) · SQL Profiler

Databases & Data Platforms

Oracle (11g/12c/19c, RAC, Data Guard) · SQL Server (2012/2014/2016/2019, Always On AG) · PostgreSQL · MySQL · Snowflake · MongoDB · Redis · Cassandra

Data Engineering & ETL

Airflow · AWS Glue · Azure Data Factory (ADF) · SSIS · Informatica

Reporting & Analytics Platforms

Snowflake · Azure Synapse Analytics · Power BI · SQL Server Reporting Services (SSRS)

Security & Compliance

TDE · Encryption at Rest / In Transit · IAM · Role-Based Access Control (RBAC) · Data Masking · HIPAA · PCI-DSS · SOC 2

Professional Experience

Senior Database Engineer,
CAI (Computer Aid, Inc.) - MedSure Health

01/2022 – Present | Remote

ClinicalOne - Enterprise platform supporting **clinical, operational, and financial data management** for **80+** healthcare organizations, focused on secure PHI processing, real-time analytics, and cloud modernization.

Key Skills: Oracle 19c RAC · Data Guard · AWS RDS (PostgreSQL, SQL Server) · Snowflake · AWS Glue · Airflow · Terraform · Liquibase · GitHub Actions · Power BI · Grafana · SolarWinds DPA · IAM · KMS · Secrets Manager · HIPAA · SOC 2

Role & Impact

- Led modernization of MedSure Health's ClinicalOne platform, a multi-tenant healthcare data system supporting **4M+** patient records.
- Delivered resilient, cloud-native data solutions that enhanced platform scalability, analytics, and compliance across clinical, financial, and operational systems.

Key Achievements

- **Enabled Zero-Downtime Modernization of Legacy Systems:** Led the migration of **Oracle 12c** environments to **Oracle 19c** with **RAC** and **Data Guard**, eliminating historical failover gaps and delivering **95% uptime** for patient-critical systems. Applied advanced Oracle **recovery strategies** and **cross-region failover** to protect against both logical and physical failures.
- **Reduced Infrastructure Costs and Improved Flexibility with Cloud Adoption:** Architected **AWS RDS (PostgreSQL, SQL Server)** solutions to offload non-critical workloads from legacy systems, reducing operating costs by **35%** and enabling on-demand scalability for microservices-based platforms integrated with **EHR systems**.
- **Transformed Data Insights through Cloud-Native Analytics:** Designed and delivered **Snowflake** data platform supporting over **10TB** of PHI and operational data. Enabled cross-departmental analytics, from clinical outcomes to operational KPIs, improving executive reporting speed **from weeks to hours**.
- **Optimized Enterprise Data Pipelines for Performance and Compliance:** Reduced batch ETL processing times by **50%** using optimized **Airflow** orchestration and **AWS Glue** for data transformation pipelines. Ensured PHI lineage, auditability, and encryption-in-motion through governance-aligned architecture.
- **Accelerated Delivery and Increased Quality through Automation:** Automated schema management and infrastructure provisioning via **Liquibase, Terraform, and GitHub Actions**, shortening deployment cycles **from weeks to hours** and reducing defect rates in data environments by **40%**.
- **Strengthened Operational Resilience through Unified Monitoring:** Integrated **Grafana, SolarWinds DPA, and AWS-native** monitoring for end-to-end observability across **Oracle, RDS, and Snowflake** environments. Reduced mean time to detect (MTTD) database incidents **from 2 hrs to under 30 mins**.

Architecture & Platform Engineering

- Architected **hybrid Oracle RAC / Data Guard** and **AWS RDS** environments supporting multi-region healthcare SaaS workloads.
- Integrated **PostgreSQL (RDS)** and **SQL Server (RDS)** to support microservices while retaining mission-critical Oracle systems for core processing.
- Built multi-environment infrastructure-as-code foundations with **Terraform** and **AWS-native** tooling.

Data Engineering & Analytics

- Delivered enterprise-grade **Snowflake** data warehouse with secure **PHI governance**, enabling near real-time insights for finance, clinical outcomes, and operational KPIs.
- Orchestrated pipelines with **Airflow, Glue, and Python**, ensuring auditable, **SLA-compliant** data flow with full encryption.
- Enabled self-service **reporting** and advanced **analytics** across **10+** business domains via **Snowflake** and **Power BI**.

DevOps & Automation

- Standardized database **deployment pipelines** via **Liquibase, Terraform, and GitHub Actions** for Oracle and AWS estates.
- Established **infrastructure-as-code patterns** with audit trails for compliance reporting and rollback capabilities.
- Delivered **automated credential management** integrated with **AWS IAM, Secrets Manager, and Oracle Wallet**.

Security & Compliance

- Enforced encryption at rest (**TDE, KMS**) and in transit (**SSL, TLS 1.2+**) across **Oracle, PostgreSQL, and Snowflake**.
- Applied **least-privilege principles** through IAM, Oracle roles, and fine-grained permissions.
- Supported annual **HIPAA, SOC 2, and GDPR** audits with **fully traceable** data operations and security controls.

Senior Database Engineer,

02/2018 – 12/2021

Opticare Health Systems

CareSphere – Cloud-based platform supporting **clinical operations, patient management, and financial analytics** for **60+ hospitals**, focused on secure PHI data management and real-time reporting through **Azure** services.

Key Skills: Azure (SQL Database, Managed Instance, Synapse Analytics, Data Factory, Data Lake Gen2) · SQL Server (Always On AG) · Oracle 12c · SSIS · Informatica · Power BI · Microsoft Purview · ARM Templates · Bicep · Azure DevOps · Azure Monitor · TDE · RBAC · HIPAA · HITRUST

Role & Impact

- Led database modernization and cloud migration for **CareSphere**, Opticare's integrated care management platform serving **60+ hospitals and clinics**.
- Delivered **Azure**-based data solutions to improve **governance, scalability, and analytics** for clinical, billing, and patient engagement workflows.

Key Achievements

- **Modernized Mission-Critical Databases for Cloud Scale:** Migrated **30+ SQL Server** to **Azure SQL Database** and **Azure Managed Instance**, reducing maintenance effort by **45%** and improving availability across **200+ clinical workflows**.
- **Enabled Real-Time Data Access for Operational Teams:** Designed a centralized data lake using **Azure Data Lake Gen2** and **Synapse Analytics**, supporting self-service reporting and analytics across scheduling, patient history, and claims operations.
- **Accelerated ETL Performance and Data Quality:** Refactored ETL pipelines using **Azure Data Factory** and SQL-based transformations, cutting data refresh lag from **6 hrs to under 1 hr** and increasing reliability for **120+ downstream reports**.
- **Improved Compliance and Access Control:** Implemented **data masking, TDE**, and fine-grained **role-based access** using **Azure Active Directory** and **SQL security policies**, aligning all systems with **HIPAA** and **HITRUST** requirements.

- **Enhanced Monitoring and Performance Tuning:** Integrated **Azure Monitor**, **Query Store**, and custom **telemetry** dashboards, reducing query-related incident volume by **60%** and enabling proactive performance management.

Architecture & Platform Engineering

- Migrated **legacy SQL Server** workloads to **Azure SQL Database** and **Azure Managed Instance** using **Database Migration Service**.
- Consolidated fragmented environments into geo-redundant, SLA-backed Azure PaaS offerings.
- Optimized storage and compute provisioning using reserved capacity, elastic pools, and scaling policies.

Data Engineering & Analytics

- Built and governed an enterprise data lake on **ADLS Gen2** for clinical and financial data domains.
- Orchestrated data pipelines with **ADF**, stored procedures, and metadata-driven ETL design for reliability.
- Delivered **Synapse-based** analytics to power **Power BI dashboards** for operations and finance leaders.

Security & Compliance

- Applied encryption at rest (**TDE**) and in transit (**TLS 1.2+**) across all Azure-hosted databases.
- Enforced access via **Azure AD integration**, dynamic data masking, and SQL row-level security.
- Supported internal audits and annual **HITRUST** certification reviews with traceable, governed data flows.

DevOps & Automation

- Standardized database deployments using **Azure DevOps** release pipelines and schema comparison tools.
- Implemented parameterized **ARM templates** and **Bicep modules** for repeatable infrastructure provisioning.
- Enabled **automated rollback** and **validation testing** for all database updates in lower and production tiers.

Database Engineer, *Liberty Finance Group*

03/2014 – 01/2018

FinSure - On-premises platform supporting **secure payment processing, loan management, and financial reporting** across **50+ institutions**, focused on **PCI-DSS** compliance, data integrity, and operational resilience.

Key Skills: Oracle RAC · SQL Server (Always On) · SSIS · Informatica · Power BI · TDE · Role-Based Access Control · PCI-DSS · PowerShell · OEM · Redgate · SQL Profiler

Role & Impact

- Supported secure, high-availability **Oracle** and **SQL Server** environments for **FinSure**, a financial transaction and reporting platform serving **50+ institutions**.
- Focused on **PCI-DSS** compliance, data integrity, reporting optimization, and operational resilience for payment, loan, and audit systems.

Key Achievements

- Built and maintained **Oracle RAC** and **SQL Server environments** for high-volume financial processing, ensuring continuous availability and compliance with PCI-DSS.
- Reduced **reporting cycles from 24 hrs to 4 hrs** by optimizing ETL pipelines (**SSIS, Informatica**), accelerating financial reconciliation and reporting.

- Improved **data quality and audit readiness** through validation frameworks, referential integrity, and audit logging, reducing discrepancies by 30%.
- Delivered **disaster recovery solutions (Oracle Data Guard, SQL Server log shipping)** achieving **RPO under 15 mins, RTO under 1 hour** for core systems.
- Automated routine maintenance, backups, and deployments via **PowerShell, SQL Agent**, and proactive monitoring tools (**OEM, Redgate, SQL Profiler**).